FYS5429/9429 Lecture April 9

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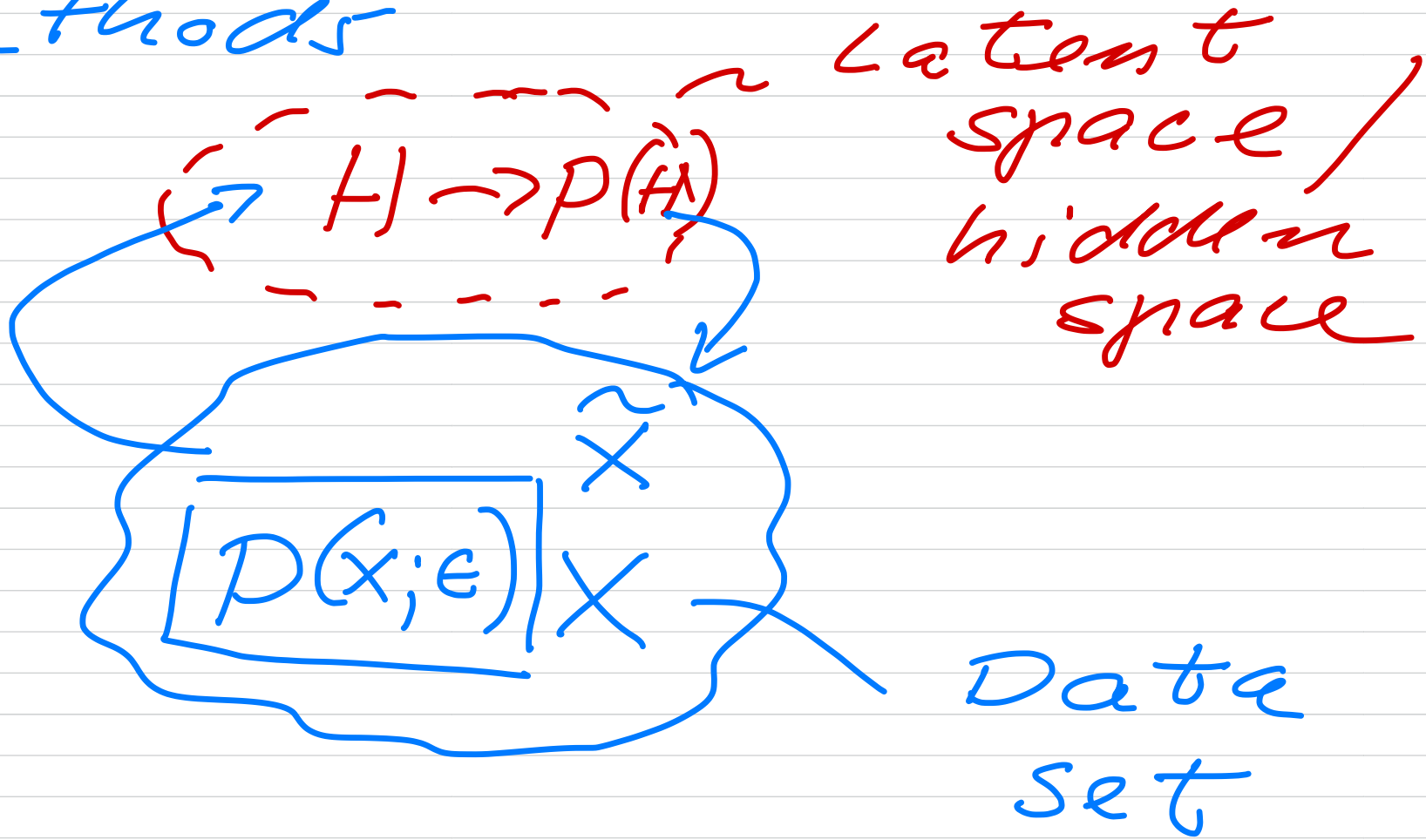
Basics of all generative methods



Θ = parameters of a generative model.

Basic statistics

Domain of stochastic variables

$$X = \{x_0, x_1 - - \quad x_{n-1}\}$$

$$Y = \{y_0, y_1 - - \quad y_{n-1}\}$$

$$P_X(X = x_i) = P(x)$$

$$P_y(Y = y_j) = P(y_j)\,(P(Y))$$

union of events

$$P(X \cup Y) = P(X) + P(Y)$$

$$- P(X \cap Y)$$

product rule

$$P(X \cup Y) = P(X, Y)$$

$$= P(X|Y)P(Y) = P(Y|X)P(X)$$

if $X \wedge Y$ are i.i.d.

$$P(X \cup Y) = P_X(X) P_Y(Y)$$

Marginal probability

$$P(X) = \sum_{i=0}^{m-1} P(X, Y = y_i)$$

$$= \sum_{i=0}^{m-1} P(X \mid Y = y_i) P(Y = y_i)$$

$$= \sum_{i} P(X \mid y_i) P(y_i)$$

Conditional probability

(if $P(y) > 0$), from product rule

$$P(x|y) = \frac{P(x,y)}{P(y)}$$

$$= \frac{P(x,y)}{\sum_i P(y|x_i)\,P(x_i)}$$

Bayes' theorem

$$P(x|y) = \frac{P(y|x)\,P(x)}{P(y)}$$

posterior

$$= \frac{P(y|x)\,P(x)}{\sum_i P(y|x=x_i)\,P(x_i)}$$

prior

likelihood

Energy models (AKA Boltzmann machines)

$$X = \{x_0, x_1 \ldots x_{m-1}\}$$

i.i.d.

$$P(x) = \prod_{x_i \in X} P(x_i)$$

Maximum likelihood $\Rightarrow$

$$p(x) \rightarrow p(x;\theta)$$

θ are the parameters

$$\hat{\theta} = \underset{\theta \in \mathbb{R}^p}{\operatorname{argmax}} \prod_i p(x_i)$$

$$= \underset{\theta \in \mathbb{R}^p}{\operatorname{argmax}} \; p(x;\theta)$$

optimization $\vec{\nabla}_\theta P(x;\theta) = 0$

equivalent statement

$$\hat{\theta} = \underset{\theta}{\arg\max} \log p(x; \theta)$$

or

$$\hat{\theta} = \underset{\theta}{\arg\min} \left(-\log p(x; \theta)\right)$$

Introduce latent space (or hidden space)

$$H = \{ h_0, h_1, \ldots, h_{m-1} \}$$

$$P(x; \theta) \rightarrow P(x, H; \theta)$$

we want the marginal probability

$$P(x; \theta) = \sum_{h_i \in H} P(x, h_i; \theta)$$

$$P(x_i, h_j; \theta) = \frac{f(x_i, h_j; \theta)}{Z(\theta)}$$

normalization constant / partition function

$$Z(\theta) = \sum_{x_i} \sum_{h_j} f(x_i, h_j; \theta)$$

Boltzmann

$$P(E_i, \beta) = \frac{e^{-\beta E_i}}{Z(\beta)}$$

Energy

inverse temp

$$Z(\beta) = \sum_{E_i} e^{-\beta E_i}$$

Example: Binary-Binary RBM

$$x_i = \{0, 1\} \qquad h_j = \{0, 1\}$$

$$f(x, h; \theta) = e^{-E(x, h; \theta)}$$

$$E(x, h; \theta) = \sum_{i=0}^{m-1} x_i a_i + \sum_{j=0}^{m-1} h_j b_j + \sum_{i,j}^{m-1\,m-1} x_i w_{ij} h_j$$

biases

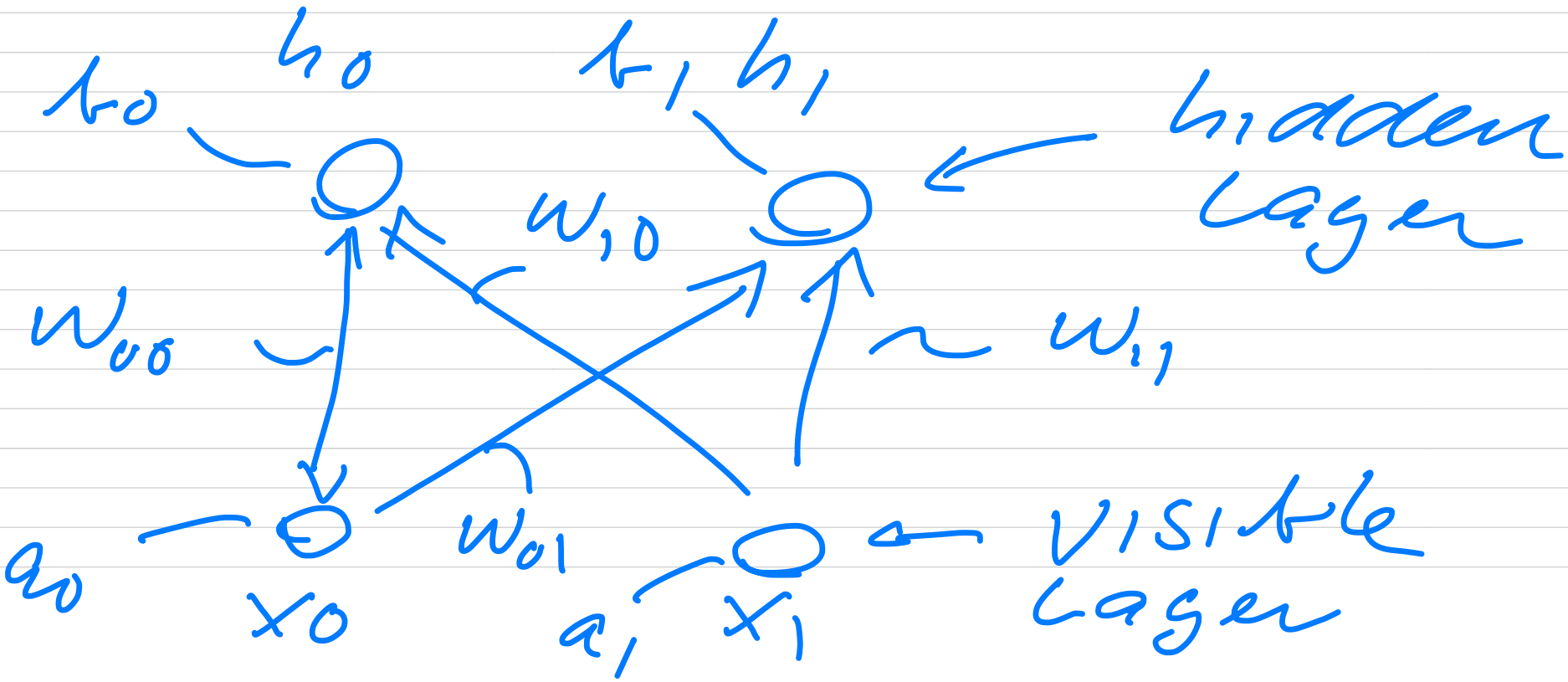
weights

$$\theta = \{ \vec{a}, \vec{b}, W \}$$

$m = 2 \quad \wedge \quad n = 2$

$h, x = \{0, 0\}, \{0, 1\}, \{1, 0\}$

$\{1, 1\}$



$$Z(\theta) = \sum_{x} \sum_{h} f(x, h; \theta)$$

$2^2 = 4$ configs

$n = m = 10 \Rightarrow 2^{10} = 1024$

$- \, ' \, - = 20 \Rightarrow 2^{20} \sim 10^7$

$$P(x;\theta) = \frac{1}{Z(\theta)} \prod_{x_i \in X} f(x_i;\theta)$$

$$= \frac{1}{Z(\theta)} \prod_{x_i} \left[\sum_{h_j} f(x_i, h_j;\theta) \right]$$

we want

$$\nabla_\theta \log P(X;\theta) = 0$$

$$= \nabla_\theta \left[\sum_{x_i} \log f(x_i;\theta) \right] -$$

$$\nabla_\theta \log Z(\theta) = \frac{\nabla_\theta Z(\theta)}{Z(\theta)}$$

$$= \frac{\nabla_\theta \left[\sum_{x_i} f(x_i ; \theta) \right]}{Z(\theta)}$$

$$= \frac{\sum_{x_i} \nabla_\theta f(x_i ; \theta)}{Z(\theta)}$$

$$f(x_i;\theta) = \exp\left[\log f(x_i;\theta)\right]$$

$$\frac{\nabla_\theta Z(\theta)}{Z(\theta)} = \frac{\sum_{x_i} \nabla_\theta \left[\exp(\log f)\right]}{Z(\theta)}$$

$$= \sum_{x_i} \frac{\boxed{\underbrace{\exp(\log f)}_{f(x_i;\theta)}}\, \nabla_\theta \log f}{Z(\theta)}$$

$$\frac{f(x_i;\theta)}{Z(\theta)} = p(x_i;\theta)$$

$$= \sum_{x_i \in X} p(x_i; \theta) \nabla_\theta \log f(x_i; \theta)$$

$$= \mathbb{E}_{x \sim X}\left[\nabla_\theta \log f(x; \theta) \right]$$

optimization of cost function

$$\nabla_\theta p(x; \theta) = 0 =$$

$$\nabla_\theta \log f(x; \theta) - \mathbb{E}_{x \sim X}\left[\nabla_\theta \log f \right]$$

$$\mathbb{E}\left[\nabla_\theta \log f(x;\theta)\right]$$

$$\simeq \frac{1}{MCS} \sum_{i=1}^{MCS} \nabla_\theta \log f(x_i;\theta)$$

Monte Carlo Markov chain samples

- Metropolis sampling
- Gibbs sampling
- Langevin — " — + other

BB-RBM

$$f(x, h; \theta) = e^{-E(x, h; \theta)}$$

$$E = a^T x + b^T x + x^T W h$$

$$= \sum_{i=0}^{n-1} a_i x_i + \sum_{j=0}^{m-1} b_j h_j$$

$$+ \sum_{i,j}^{n-1, m-1} x_i w_{ij} h_j$$

$$\nabla_\theta \Rightarrow \nabla_a, \nabla_b, \nabla_W$$

Basics of Markov chain

$$P_i(t) = \sum_j W(j \to i)\, P_j(t-1)$$

transition prob

our prob. model

For discrete case, W is a so-called stochastic matrix

$$\sum_j W_{ij} = \underline{1}$$

<u>Example</u>

$$W = \begin{bmatrix} 1/2 & 1/3 & 1/3 \\ 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/3 \end{bmatrix}$$

Largest eigenvalue $= \underline{1}$

Markov chain

$$\vec{p}(t) = W \cdot \vec{p}(t-1)$$

when equilibrium has be reached $\vec{p}(t)$ is independent of t

$$\boxed{\vec{p} = W \cdot \vec{p}}$$

$(A\vec{x} = \lambda \vec{x})$

eigenvalue $= 1$, which is the largest eigenvalue of W